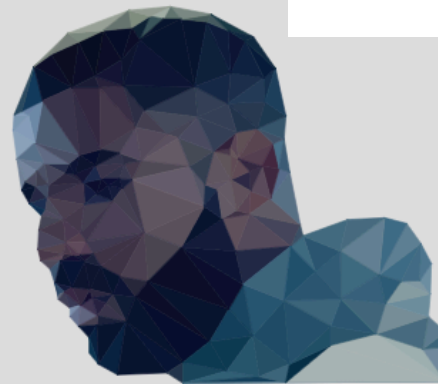


The art of information



Class 3: 01/07 (Compression 1)

ThePudding



AN EXERCISE IN LANGUAGE COMPRESSION

Are Pop Lyrics Getting More Repetitive?

By [Colin Morris](#)

Pitchfork

RESONANT FREQUENCY

Ten Meditations on Repetition in Music, With Soundtrack

In listening, thinking, and writing about music over the past eight years, I've never been able ...

By Mark Richardson

May 4, 2006

```
1 sound(["bd*4, [- cp]*2, [- hh]*4").bank("RolandTR909")
```

welcome patterns sounds reference console settings

welcome

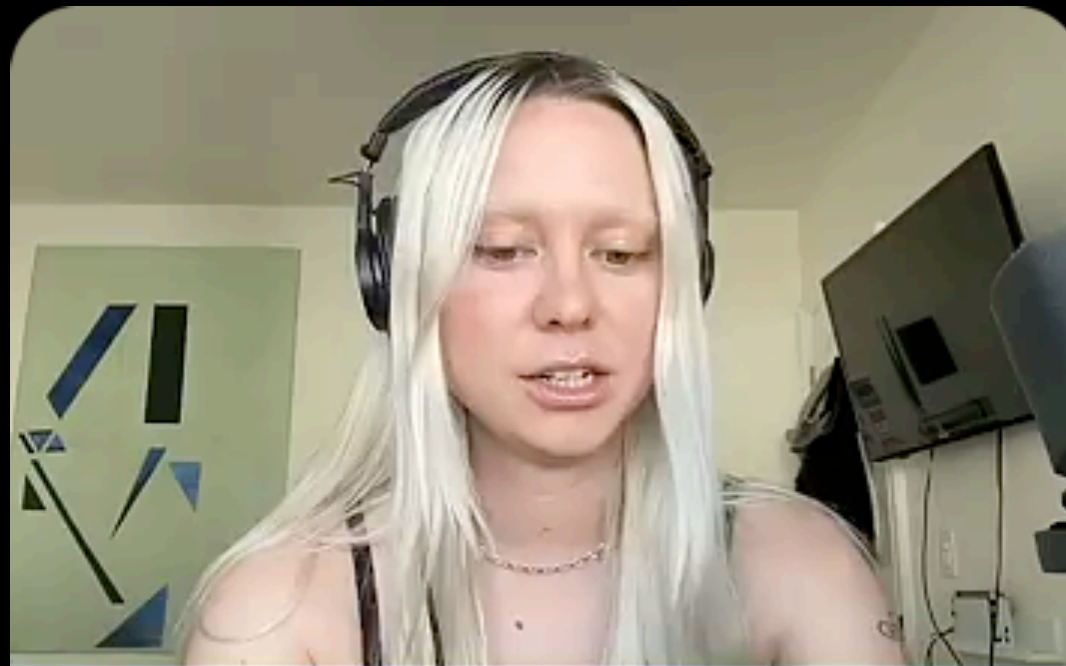
You have found [strudel](#), a new live coding platform to write dynamic music pieces in the browser! It is free and open-source and made for beginners and experts alike. To get started:

1. [hit play](#) - 2. [change something](#) - 3. [hit update](#)

To get started, check out the [interactive tutorial](#). Also feel free to join the [discord channel](#) to ask any questions, give feedback or just say hello.

about

strudel is a JavaScript version of [tidalcycles](#), which is a popular live coding language for music, written in Haskell. Strudel is free/open source software: you can redistribute and/or modify it under the terms of the [GNU Affero General Public License](#). You can find the source code at [codeberg](#). You can also find [licensing info](#) for the default sound banks there. Please consider to [support this project](#) to ensure ongoing development 💖



```

const Structures = [
  "-",
  "x",
  "[x -19 x -15 x - x -17 x -13 < - x > -]x16",
]

const gooo = 1
// off/on

baseline: note("eb1, eb2):16 [f2, f1]:16 [g2, g1]:16 [f3, f4]:16 [bb2, bb1]:16")
  .sound("supersaw")
  .slow(8)
  .postgain(2)
  .room(0.6)
  .lpf(slider(725, 300, 2000))
  .room(0.4)
  .postgain(pick(gainnn, gooo))
  .punchcard((height:200, width:1670))

const arpeggiator = [
  "[d4 bb3 eb3 d3 bb2 eb2]x16",
  "[c4 bb3 f3 c3 bb2 f2]x16",
  "[d4 bb3 g3 d3 bb2 g2]x16",
  "[c4 bb3 f3 c3 bb2 f2]x16",
]

main_arp: note(pick(arpeggiator, "c4 1 2 3").slow(2))//.rev()
  .sound("supersaw")
  .lpf(300)
  .sustain(0.5).release(0.01).attack(0)
  .room(0.6)
  .lpnv(slider(2.39525, 1.25, 6))
  .postgain(pick(gainnn, gooo))
  .punchcard((height:200, width:1670))

drums: stack(
  s("tech:5").postgain(6).pcurve(2).pdec(1).struct(pick(Structures, gooo)),
  // s("r - rim - cp - rim - cp - < rim - >2]b8").bank("KorgDOM10, dmx").speed(1.2).fast(2).post
  // s("sh").struct("[x13 -12 x10 -]").postgain(0.5).lpf(7000).bank("RolandTR808").speed(0.8).jux(rev).room(sine.
  // s("hh").struct("x*16").postgain(0.5).bank("RolandTR808").speed(1).jux(rev).room(sine.range(0.1, 0.4)).gain(
  // s("b").bank("RolandTR808").speed(0.5).room(0.5).gain(0.5).fast(2)

```



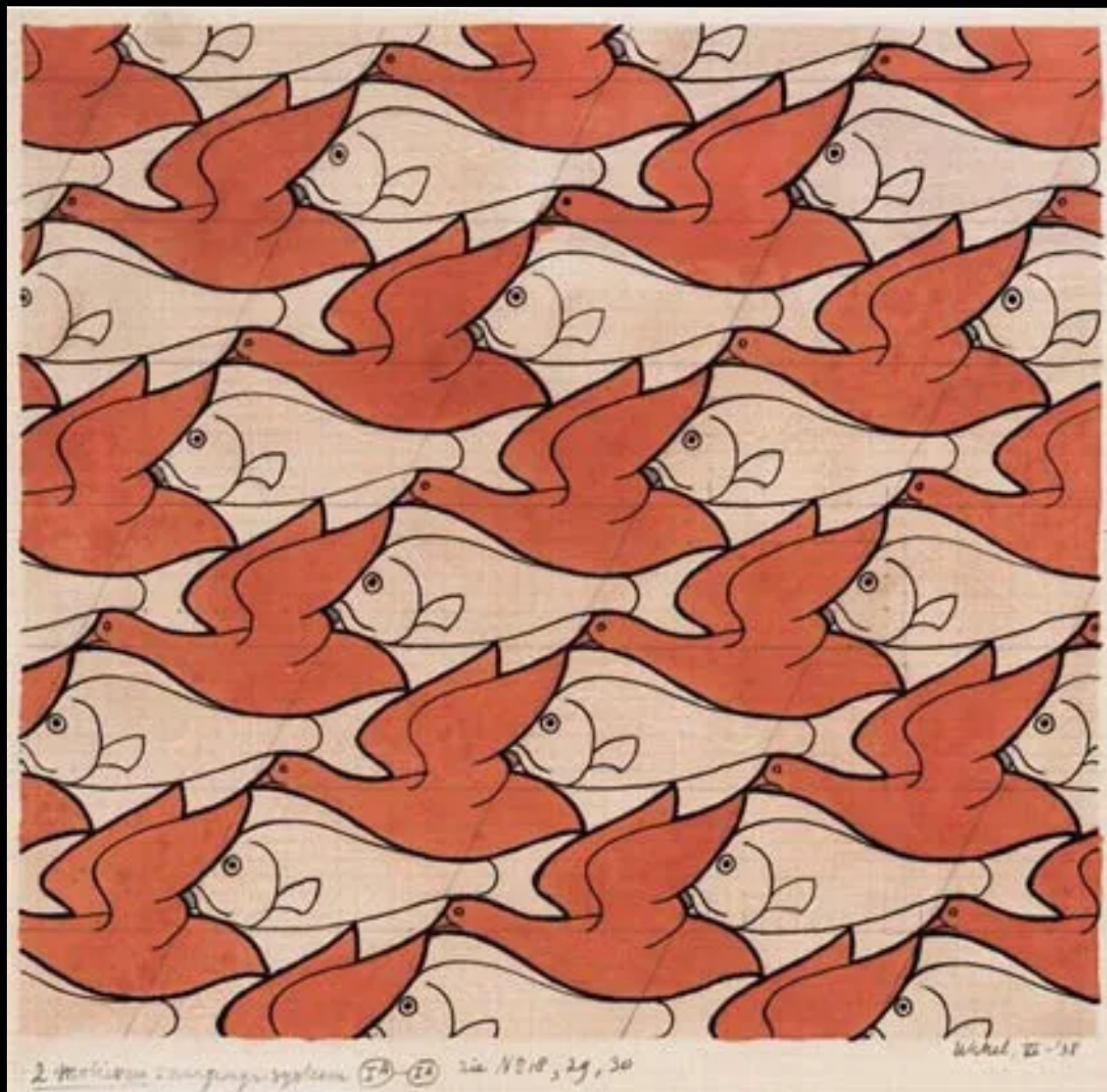
```

1 // "coastline" @by eddyflux
2 // @version 1.0
3 samples('github:eddyflux/crate')
4 setcps(.75)
5 let chords = chord("<Bbm9 Fm9>/4").dict('ireal')
6 stack(
7   stack( // DRUMS
8     s("bd").struct("<[x*<1 2> [~@3 x]] x>"),
9     s("~ [rim, sd:<2 3>]").room("<0 .2>"),
10    n("[0 <1 3>]*<2!3 4>").s("hh"),
11    s("rd:<1!3 2>*2").mask("<0 0 1 1>/16").gain(.5)
12  ).bank('crate')
13  .mask("<[0 1] 1 1 1>/16".early(.5))
14  , // CHORDS
15  chords.offset(-1).voicing().s("gm_epiano1:1")
16  .phaser(4).room(.5)
17  , // MELODY
18  n("<0!3 1*2>").set(chords).mode("root:g2")
19  .voicing().s("gm_acoustic_bass"),
20  chords.n("[0 <4 3 <2 5>>*2](<3 5>,8)")
21  .anchor("D5").voicing()
22  .segment(4).clip(rand.range(.4,.8))
23  .room(.75).shape(.3).delay(.25)
24  .fm(sine.range(3,8).slow(8))
25  .lpf(sine.range(500,1000).slow(8)).lpq(5)
26  .rarely(ply("2")).chunk(4, fast(2))
27  .gain(perlin.range(.6, .9))
28  .mask("<0 1 1 0>/16")
29 )
30 .late("[0 .01]*4").late("[0 .01]*2").size(4)

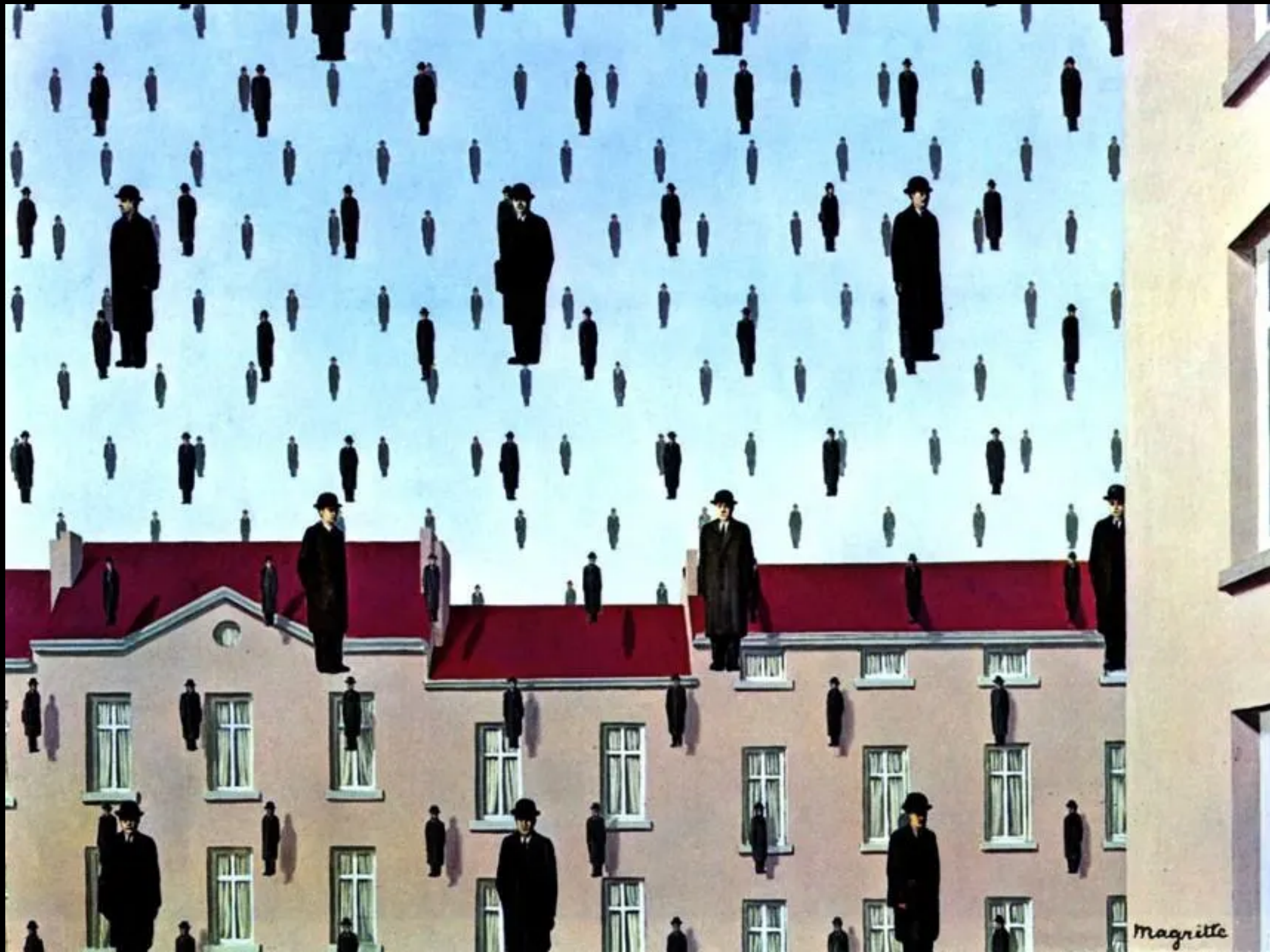
```



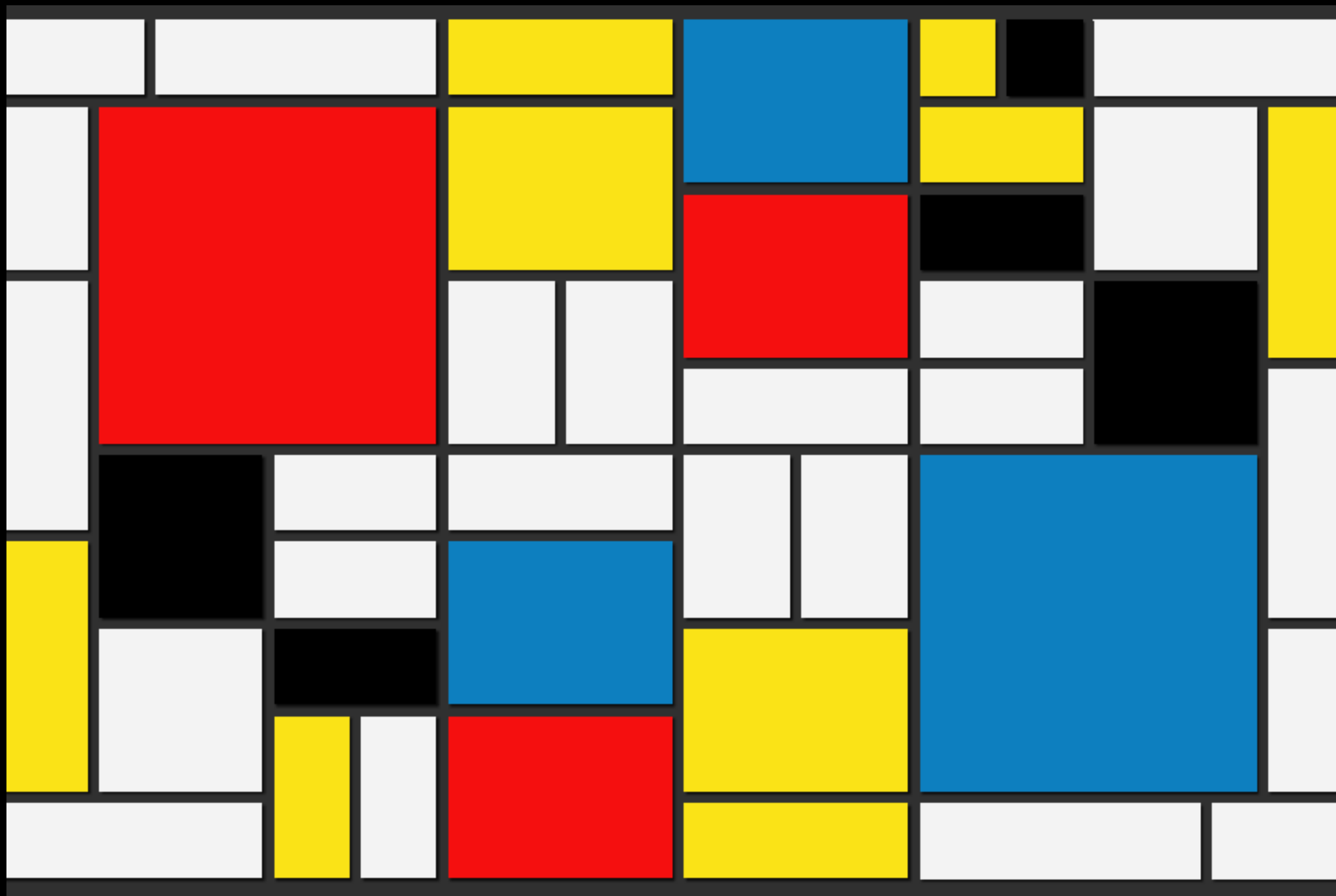

Andy Warhol



MC Escher



Rene Magritte



Piet Mondrian

but why?

The file you are trying to send exceeds the 25MB attachment limit.

But don't worry, you can send it using Google Drive.

[Send using Google Drive](#)

[Not now](#)





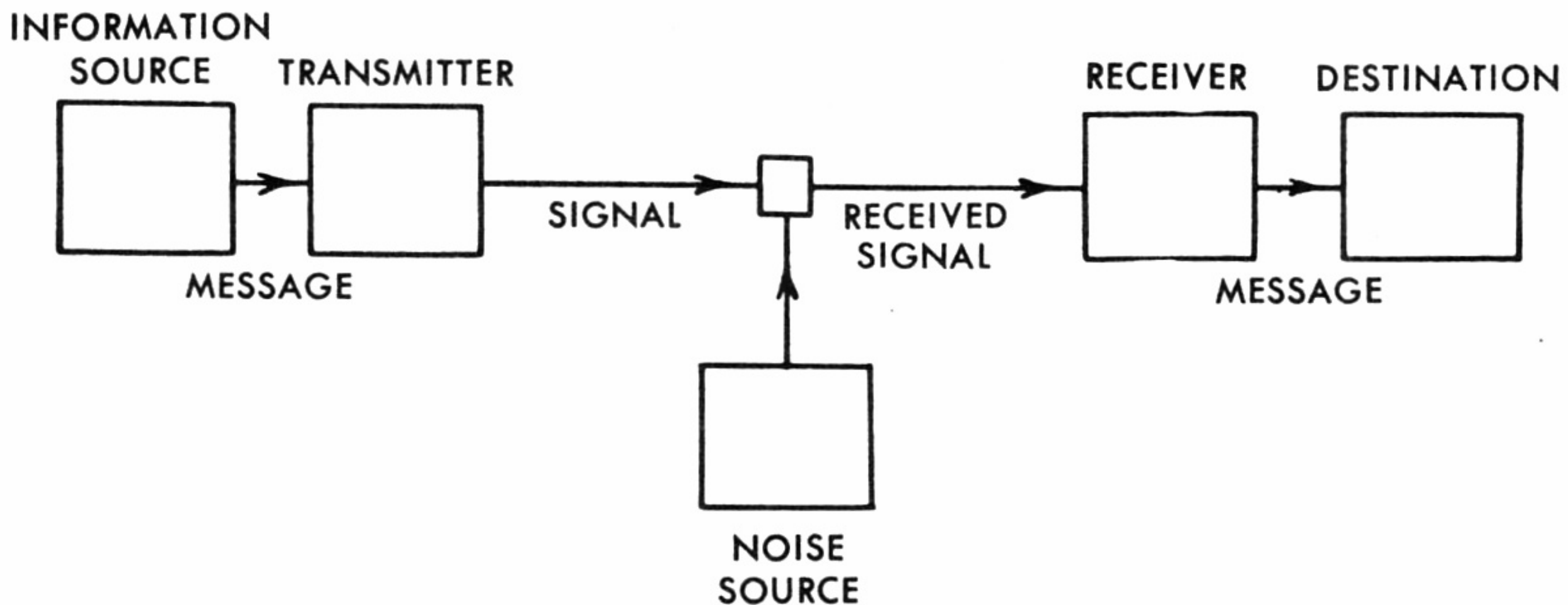


Fig. 1. — Schematic diagram of a general communication system.

lossy vs. lossless

what's the difference?

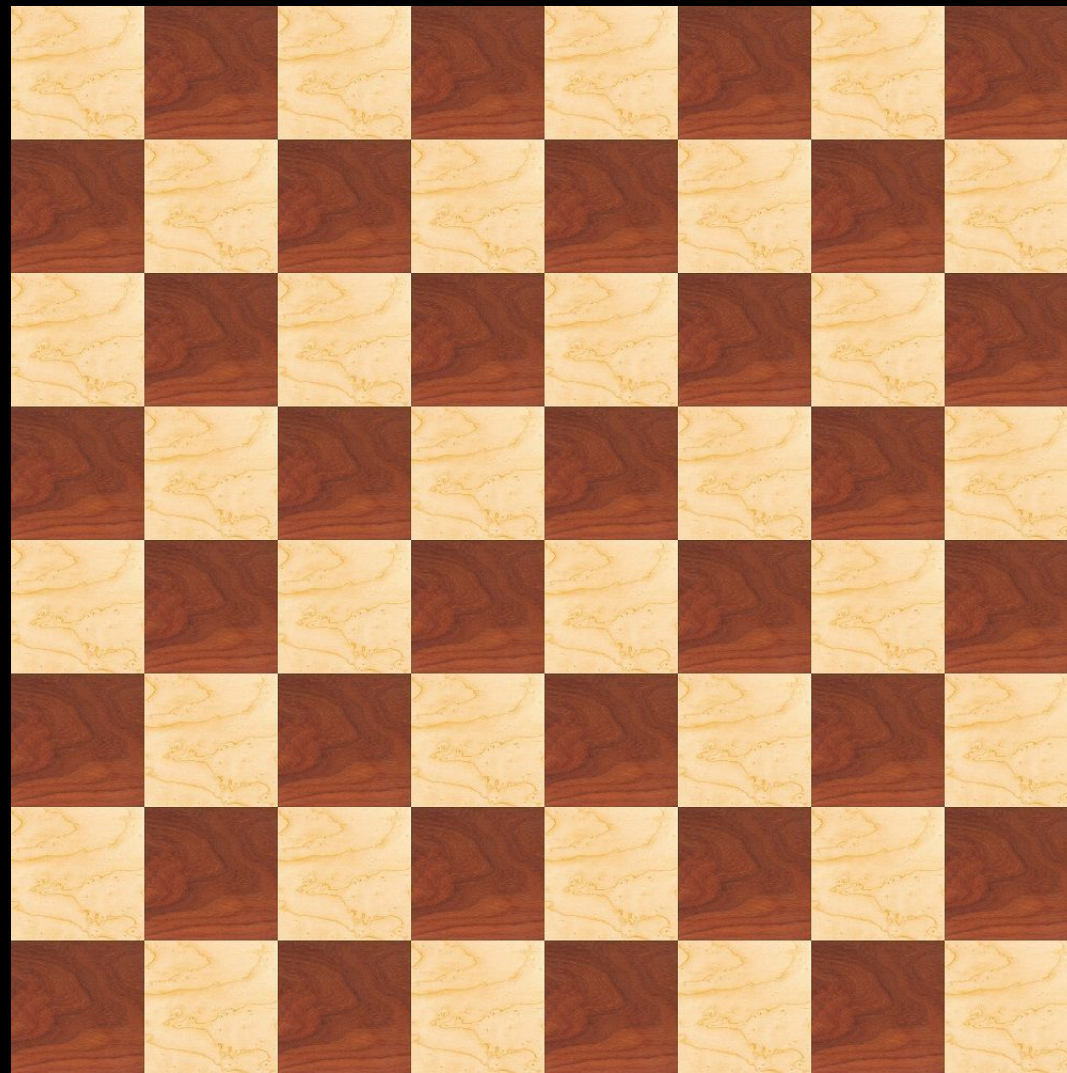


what's the difference?



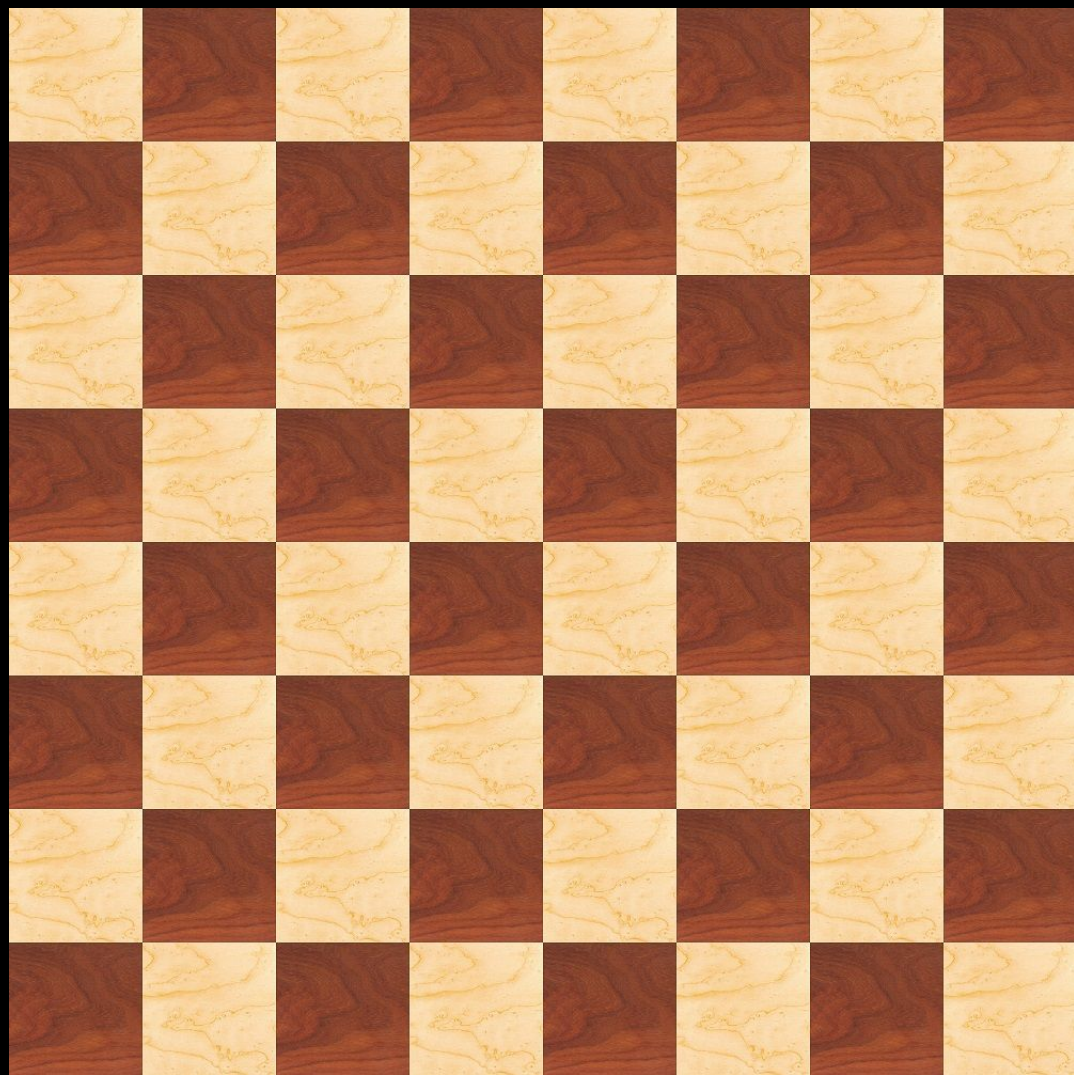
lossy: information is lost, but that
information doesn't matter much

how to compress?

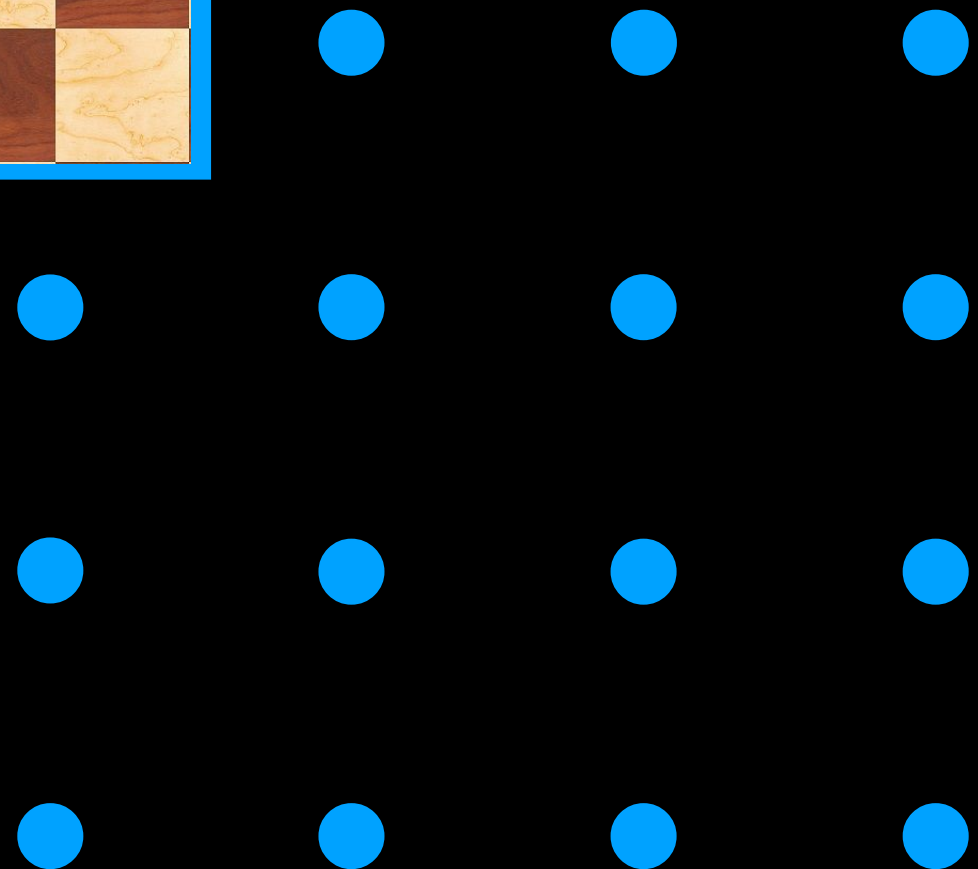


2500 pixels

how to compress?

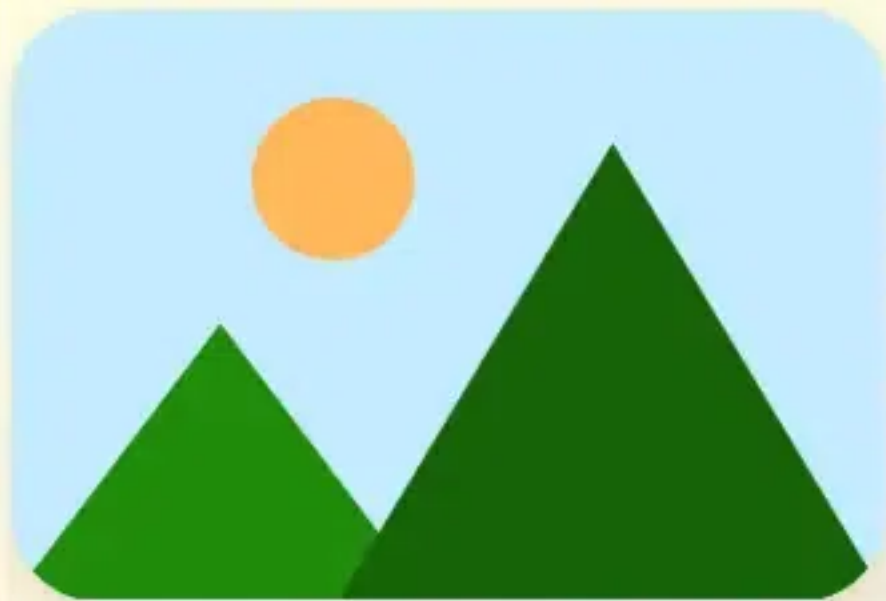


2500 pixels



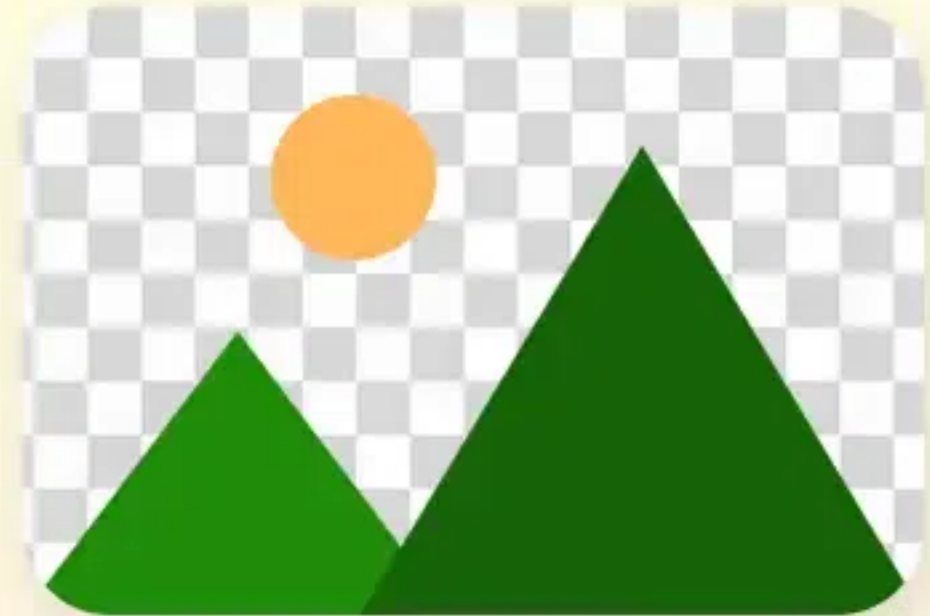
200 pixels

JPEG



vs

PNG



TOP 4 IMAGE FORMATS FOR THE WEB [2018]

JPG

GIF

PNG

SVG

VECTOR



RASTER



TRANSPARENCY



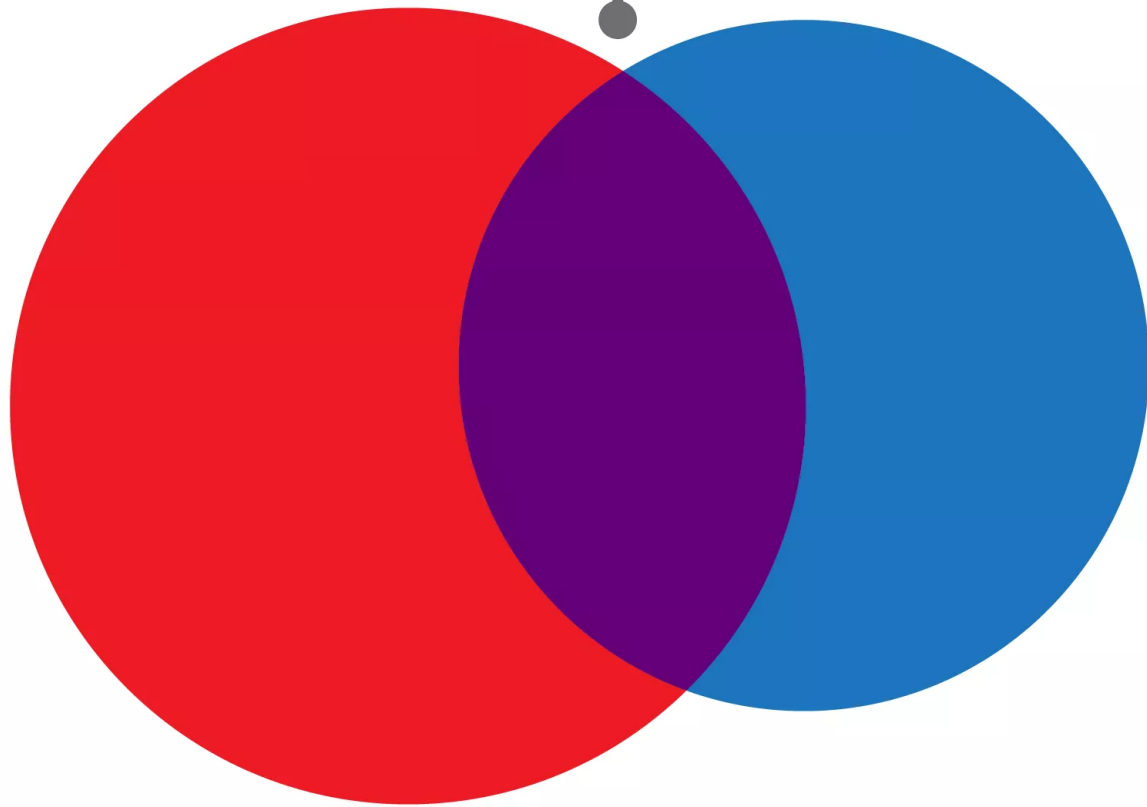
ANIMATION



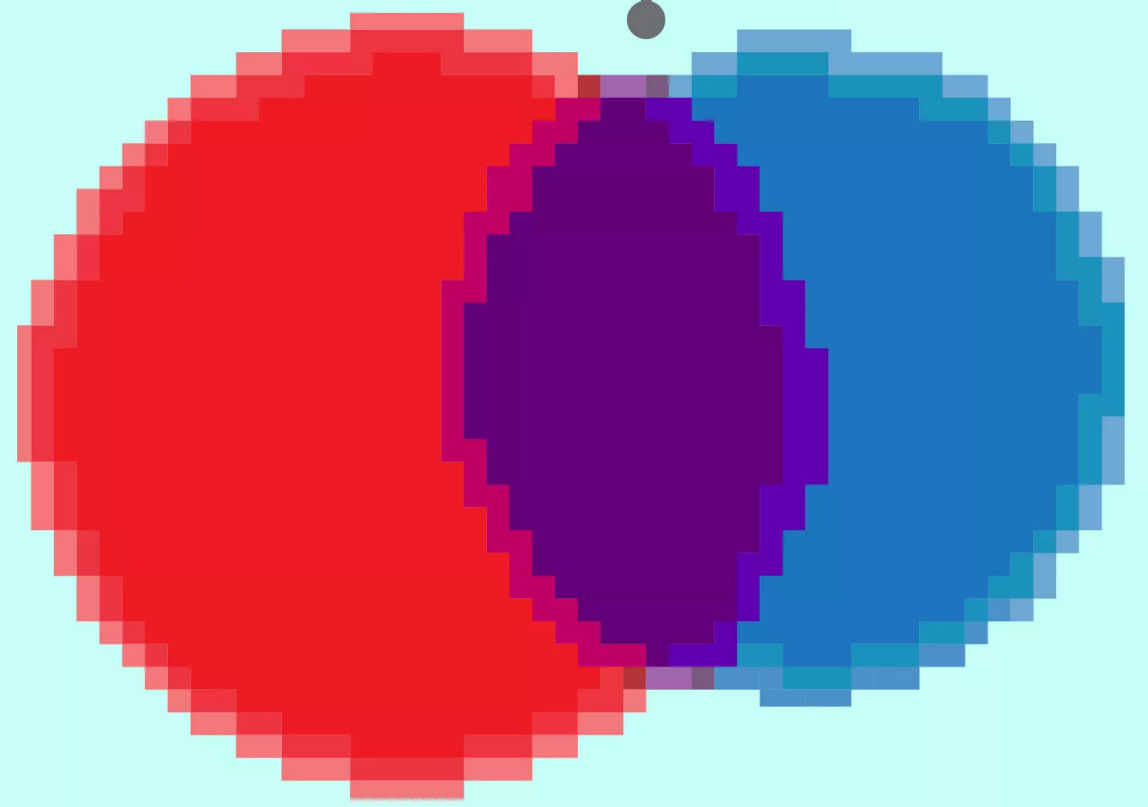
LOSSY



Vector



Raster



AI

EPS

CGM

PDF

SVG

CDR

BMP

TIFF

PCX

GIF

PNG

JPEG

format

The Language of Thought

JERRY A. FODOR

The Language and Thought Series

Jerrold J. Katz

D. Terence Langendoen

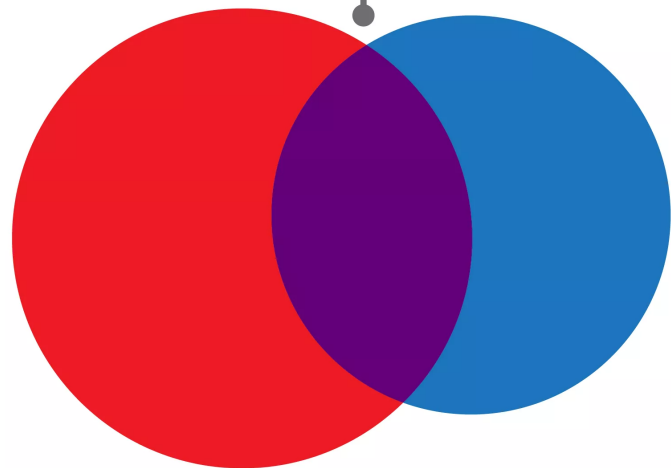
George A. Miller

SERIES EDITORS

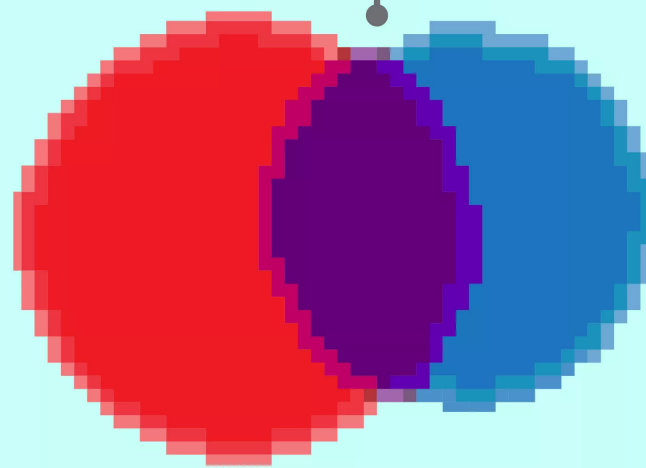
images,
but also thought

what does this get us?

Vector



Raster



AI

EPS

CGM

PDF

SVG

CDR

BMP

TIFF

PCX

GIF

PNG

JPEG

The Language of Thought

JERRY A. FODOR

The Language and Thought Series

Jerrold J. Katz

D. Terence Langendoen

George A. Miller

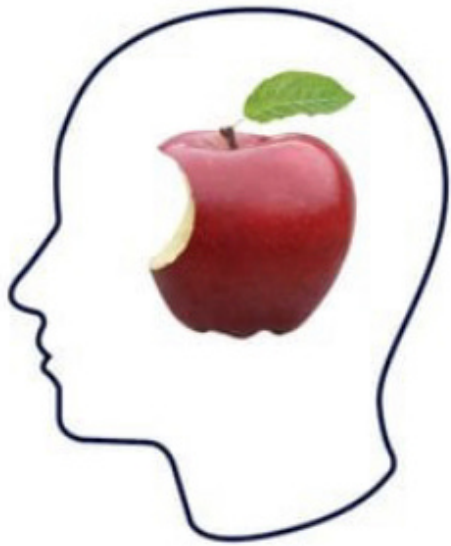
SERIES EDITORS

tiger

vs.



Imagination Spectrum



5



4



3



2



1

Return of the mental image: are there really pictures in the brain?

Zenon Pylyshyn

Rutgers Center for Cognitive Science, Rutgers University, New Brunswick, NJ 08903, USA

Psychological Review
1981, Vol. 88, No. 1, 46–66

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0033-295X/81/8801-0046\$00.75

The Medium and the Message in Mental Imagery: A Theory

Stephen Michael Kosslyn
Harvard University

format & compression



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journal homepage: www.elsevier.com/locate/COGNIT



What's magic about magic numbers? Chunking and data compression in short-term memory

Fabien Mathy^{a,*}, Jacob Feldman^b

^a *Université de Franche-Comté, 30-32 rue Mégevand, 25030 Besançon Cedex, France*

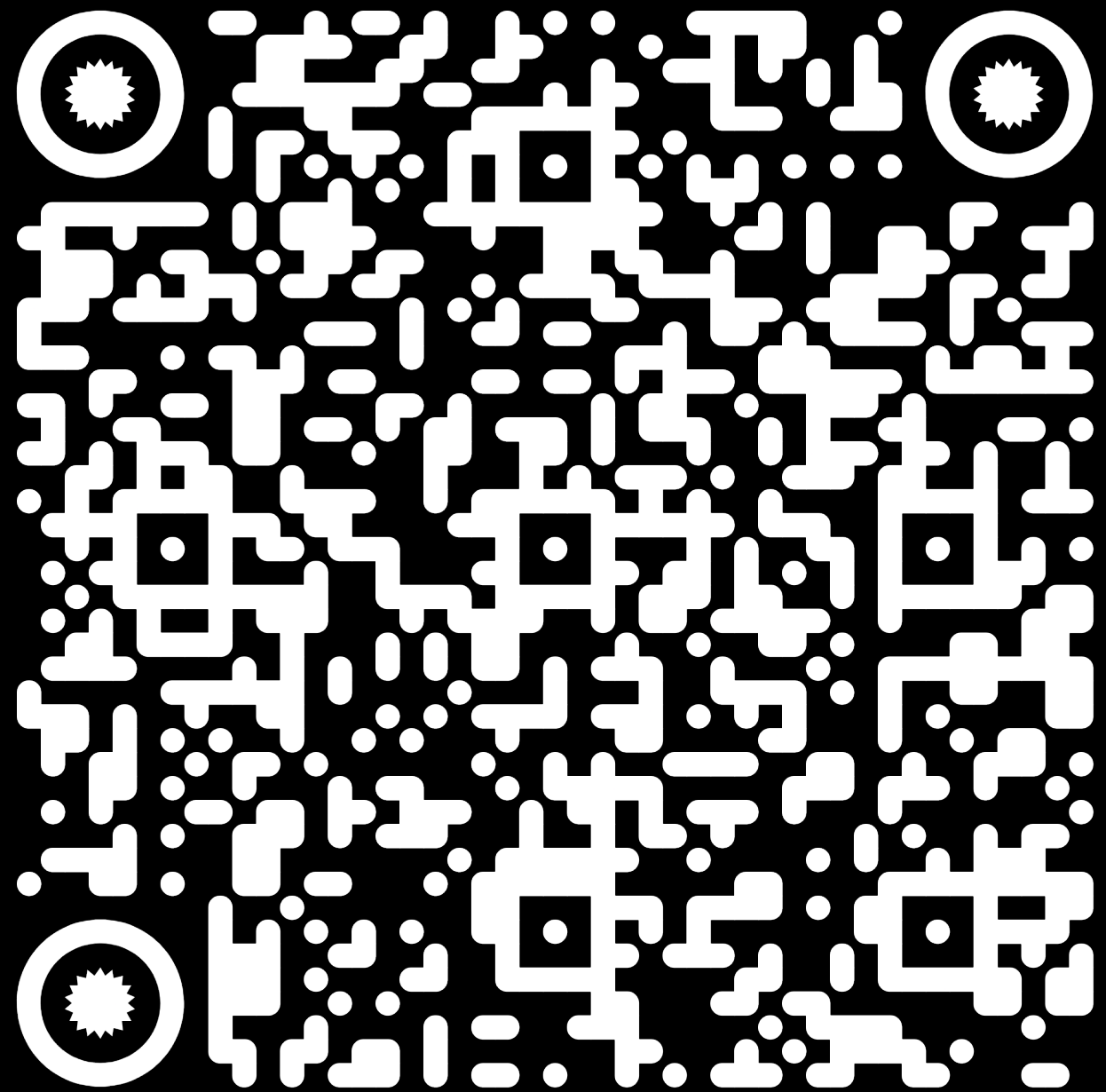
^b *Rutgers University, New Brunswick, USA*

tomorrow

least
compressible



most
compressible



The art of information



Class 3: 01/07 (Compression 1)